



2025–2026 Undergraduate Catalog

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Environmental Spatial Analysis (ESA) (B.S.)

**Location(s):** [Gainesville](#)[← Return to: Programs](#)**120 hours**

The Bachelor of Science in Environmental Spatial Analysis is a blend of both education and preparation designed to produce graduates with skills that meet workforce needs in the Geospatial Science and Technology field with little or no further preparation by employers. The Environmental and Spatial Analysis (ESA) degree is composed of a focused curriculum built around advanced Geographic Science and Technology and includes the opportunity for students to focus their learning experience through directed elective coursework in computing and information technology, earth systems, engineering, environmental science, environmental studies, geoscience, and other areas.

- Fulfill all UNG policies and graduation requirements.
- GISC 2011/L and ENST 2030 should be completed in the Core IMPACTS or as electives.
- With approval, MATH 1401 or MATH 3350 coupled with MATH 1113 or higher may substitute for GISC 3337K. Course credit can only be counted once toward degree requirements.

- Certain courses of focus areas, minors, certificates may have prerequisites that should be completed as Core classes or as Electives. Please see an advisor.
- Grades of C or higher are required for all GISC and ENVE courses.
- With approval, ENGR 1101 Intro to Engineering and Design may substitute for ENVE 1103.
- With approval, ENGR 1770 Engineering Graphics and Design may substitute for ENVE 2771K. Course credit can only be counted once toward degree requirements.
- Students that have previously completed a bachelor's degree or higher should meet with the Dean of the Lewis F. Rogers Institute for Environmental and Spatial Analysis (IESA) to discuss the academic requirements for completing the ESA degree program. Students with a B.S. degree or higher may apply courses taken at another institution but not offered by UNG. Some of these courses may be substituted for courses in one of the emphasis areas. These substitutions must be approved by the academic advisor and the Dean of IESA.
- Only three hours of special topics, service learning or internship may count toward Directed Electives.

Core IMPACTS – 42 hours

Refer to Core IMPACTS

Institution: No exceptions

Mathematics: MATH 1111, MATH 1113, or MATH 1450

Citizenship: No exceptions

Humanities: No exceptions

Writing: No exceptions

STEM: No exceptions

Social Sciences: No exceptions

Note: Please see an adviser to optimize Core IMPACTS selections and scheduling

Field of Study – 18 hours

- ENVE 1103 – Introduction to Environmental Professions
- ENVE 2221K – Surveying I
- ENVE 2771K – Graphics and Information Visualization

Select one of the following:

- ENVS 2111K – Physical Environmental Science

- ENVS 2112K – Environmental Management & Sustainability.

Select 3 hours from the following:

- CHEM 2734 – Analytical Chemistry.
- CHEM 2734L – Analytical Chemistry Laboratory.
- ENST 2030 – Environmental Studies & Sustainability.
- GEOG 1101 – Introduction to Human Geography.
- GEOG 1102 – World Regional Geography.
- GEOG 1111K – Introduction to Physical Geography.
- GEOG 1112 – Introduction to Weather and Climate
- GEOG 1112L – Introduction to Weather and Climate Lab
- GEOG 1530K – Introduction to Oceanography.
- GEOL 1101K – Survey of Environmental Geohazards
- GEOL 1121K – Physical Geology.
- GEOL 1122K – Historical Geology.
- GEOL 1530K – Introduction to Oceanography.
- GISC 2011 – Geographic Information Science
- GISC 2011L – Geographic Information Science Lab
- MATH 1401 – Elementary Statistics
- MATH 1450 – Calculus I
- MATH 2460 – Calculus II
- SCTS 2200 – Introduction to Space Studies

Major Requirements – 60 hours

Required Courses - 28 hours

- ENVE 3001K – Land Use and Conservation
- GISC 3011K – Cartography and Earth Measurement
- GISC 3012K – Global Navigation Satellite Systems
- GISC 3021K – Geospatial Science and Technology.
- GISC 3200K – Programming for Geospatial Science and Technology.
- GISC 3337K – Mathematical Methods for Geospatial Science and Technology.
- GISC 4011K – Data Acquisition & Management
- GISC 4350K – Remote Sensing of the Environment
- GISC 4470K – Spatial Analysis

Directed Electives - Select at least 15 hours

Select additional 3000/4000 level classes from: ENST, ENVE, ENVS, GEOG, GEOL, GISC, SCTS Courses*

***Note:** Some courses may have prerequisites which should be completed as Core Classes or Electives.

***Note:** Only three hours of special topics, service learning or internship may count toward Directed Electives.

Free Electives - To Bring Total Credits to 120 Hours

Minor, Certificate, or Focus Area suggested

Note: A minimum 39 credit hours at the 3000/4000 level are required for graduation.

Other Requirements

- [US and Georgia History Requirements](#)
- [US and Georgia Constitution Requirements](#)
- [Regents' Skills Program Requirements](#)

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